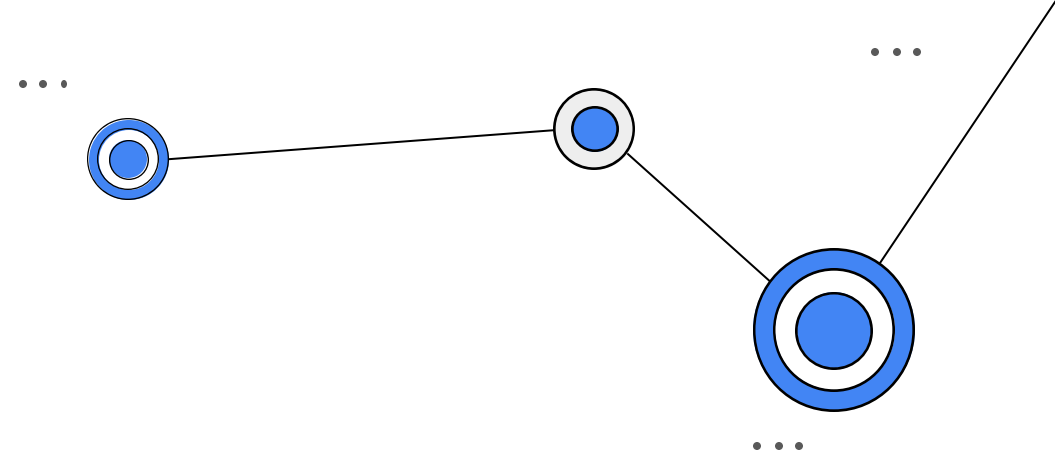
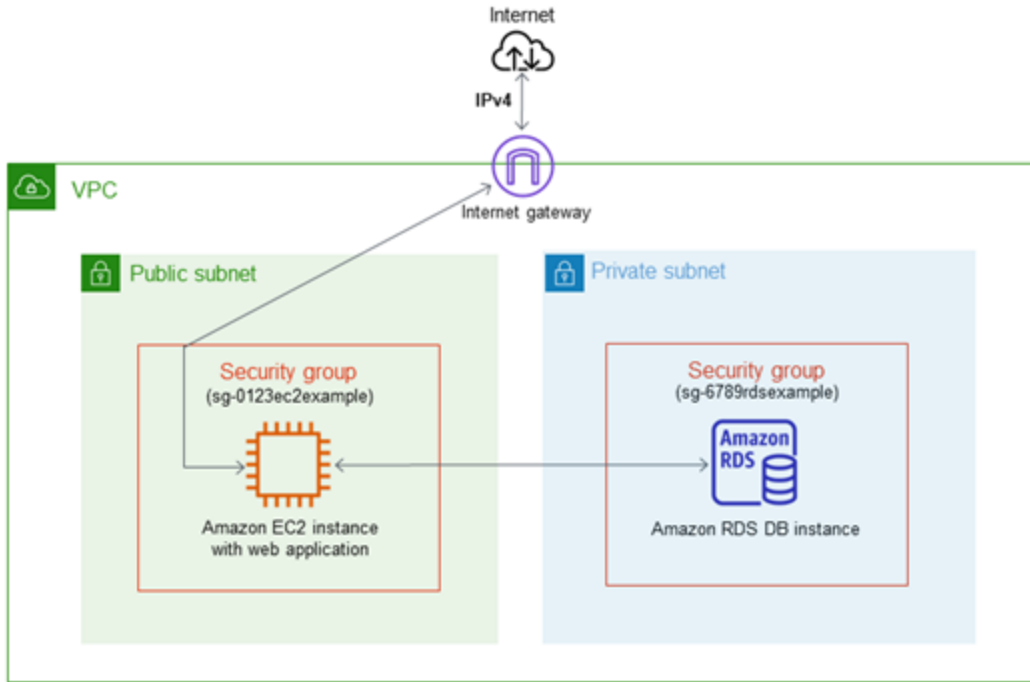
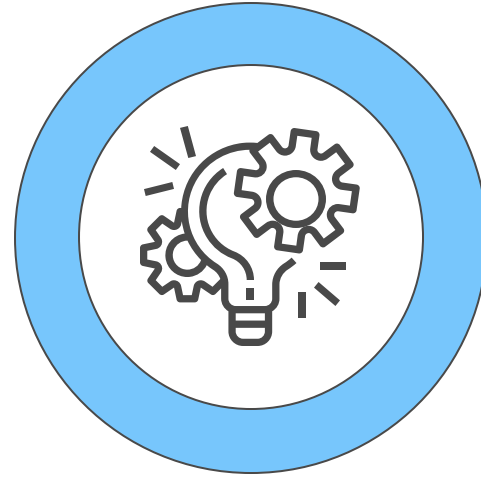




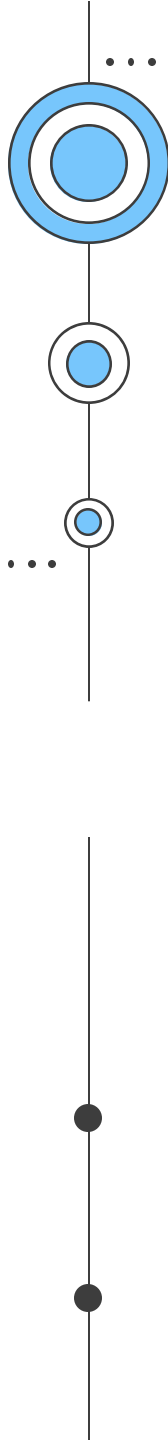
DNS, Nginx y Deployment

IIC2173 – 2026.1
Matías Araos y Daniel Salgado



Dudas

Programa
Entrega 0
(dudas muy específicas al final)
Curso
Etc
...



00

RabbitMQ

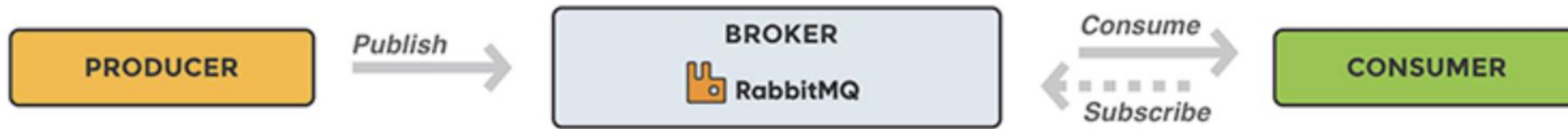
Funcionamiento básico



RabbitMQ es un software de gestión de colas de mensajes, también conocido como *intermediario de mensajes* o *gestor de colas*.

[Documentación RabbitMQ](#)





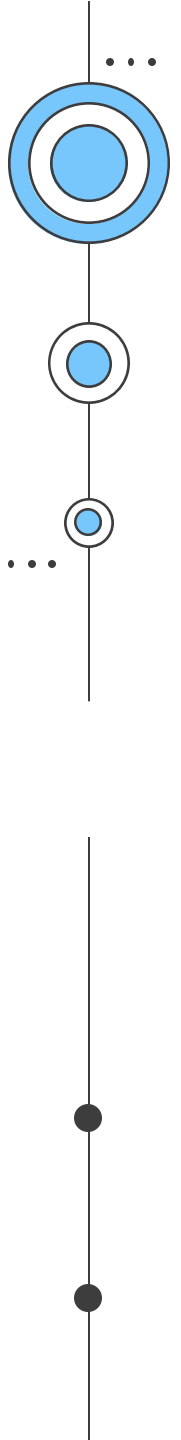
RabbitMQ permite comunicar sistemas de forma asíncrona.

Los sistemas no tienen que hablarse al mismo tiempo ni esperar respuesta inmediata.

Ventajas:

- baja acoplamiento (menor dependencia)
- evita depender de respuestas inmediatas
- permite procesar tareas en segundo plano
- ayuda a escalar sistemas con muchos eventos

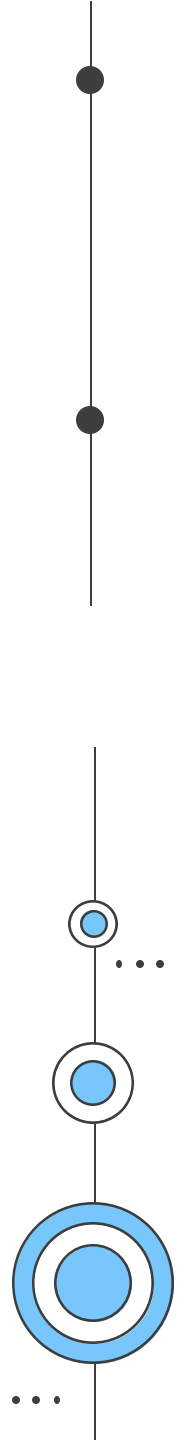
[Documentación RabbitMQ](#)



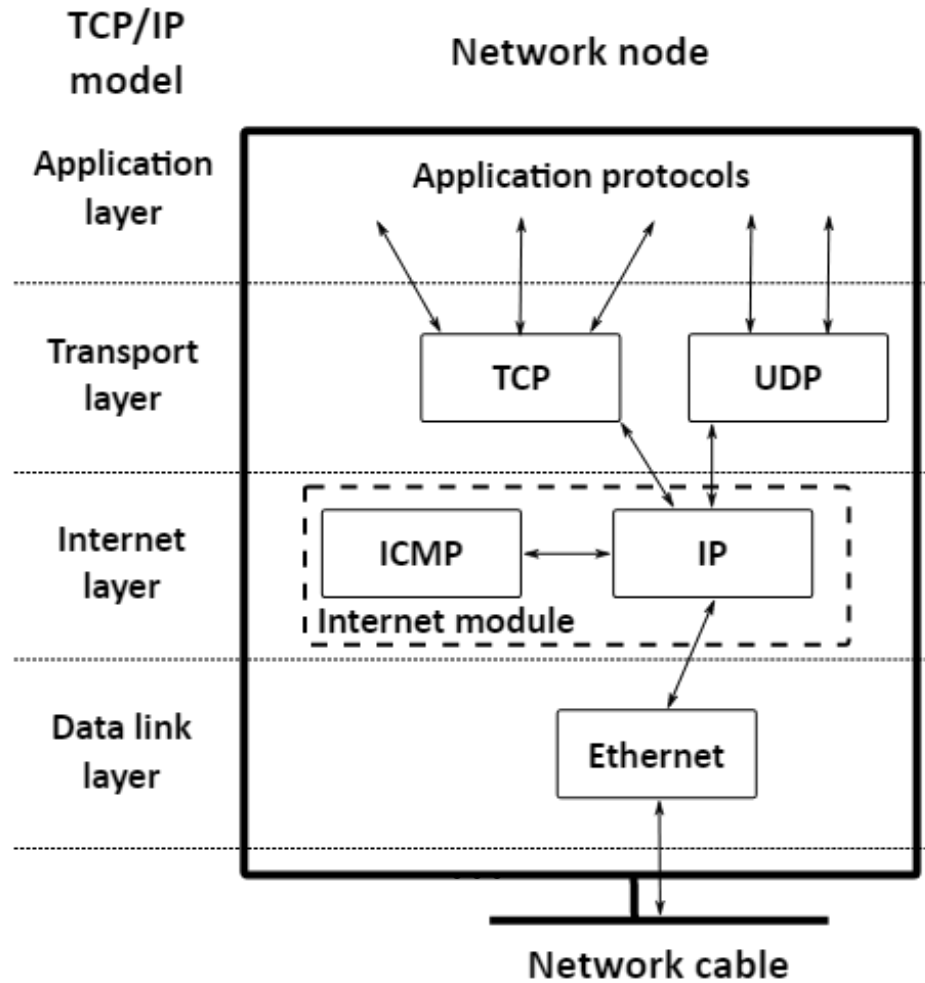
01

Internet y DNS

Funcionamiento básico

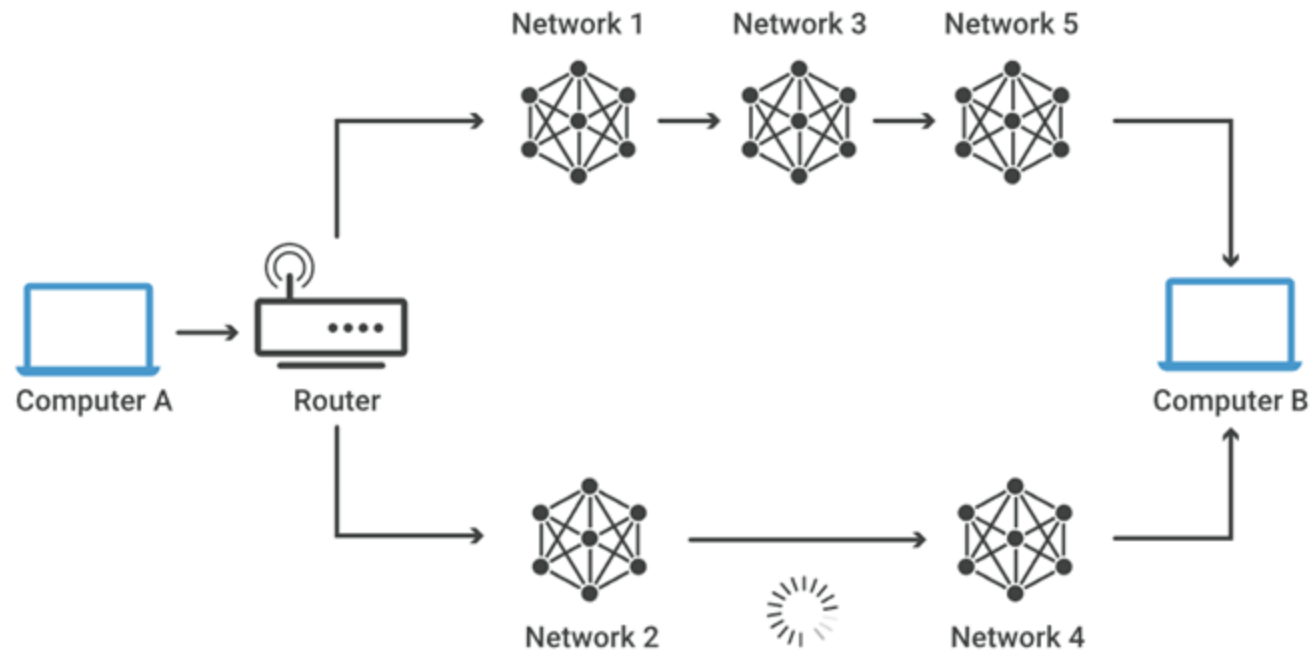


Layers de la internet



Routing en la internet

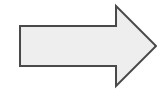
Enrutamiento



DNS

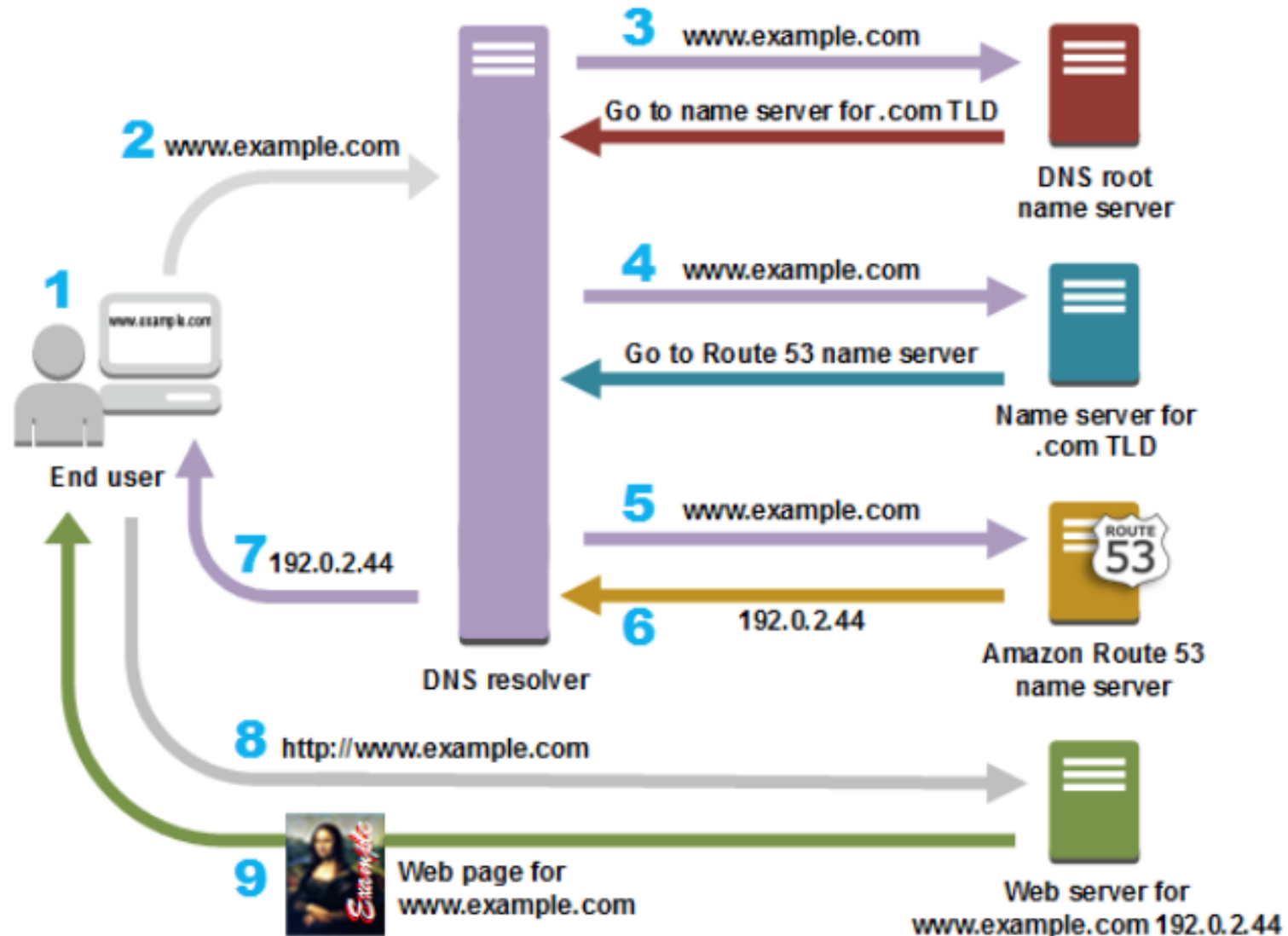
URL

Uniform Resource Locator



112.25.157.245

Resolución DNS



Registros clave

01
...

A
address - Traduce una dirección IP (v4)

02
...

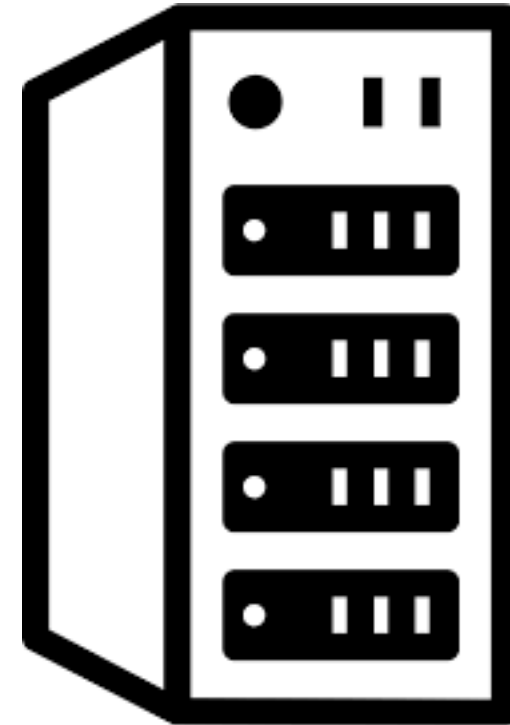
CNAME
canonical name - Traducen un nombre a otro

03
...

TXT
text - Información arbitraria, usada en validaciones

04
...

NS
nameserver - ¿Dónde puedo ir a preguntar por los registros de este dominio?

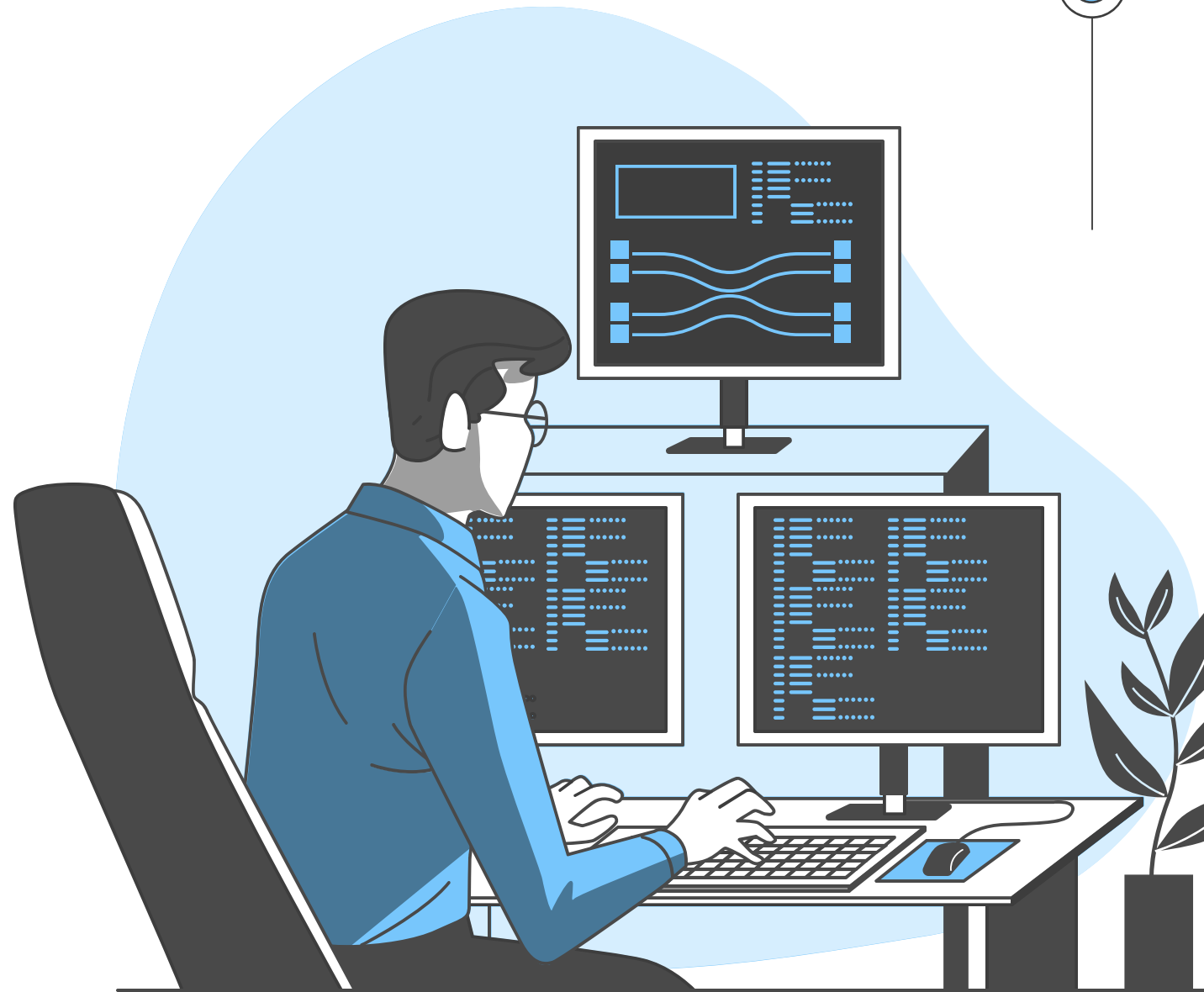


Vamos a buscar dominios!

(deben tener GitHub student pack para obtener gratuitos)

nc.me (Namecheap)

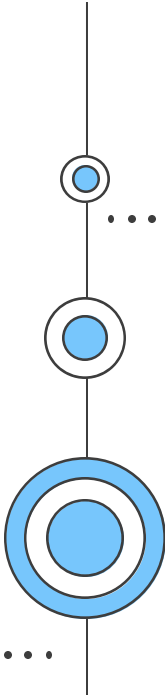
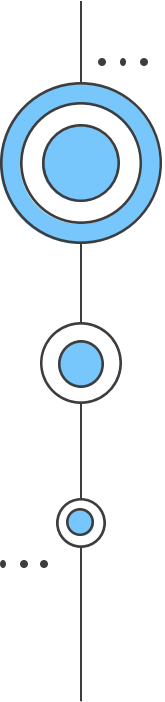
get.tech



Recapitulemos los pasos



- Levantar un servidor
- Asociar una IP elástica al servidor
- Contratar un dominio
- Configurar el dominio para que su registro A apunte a la IP elástica
- Configurar el dominio para que use un subdominio www. (CName)
- Esperar a que se propague y probar

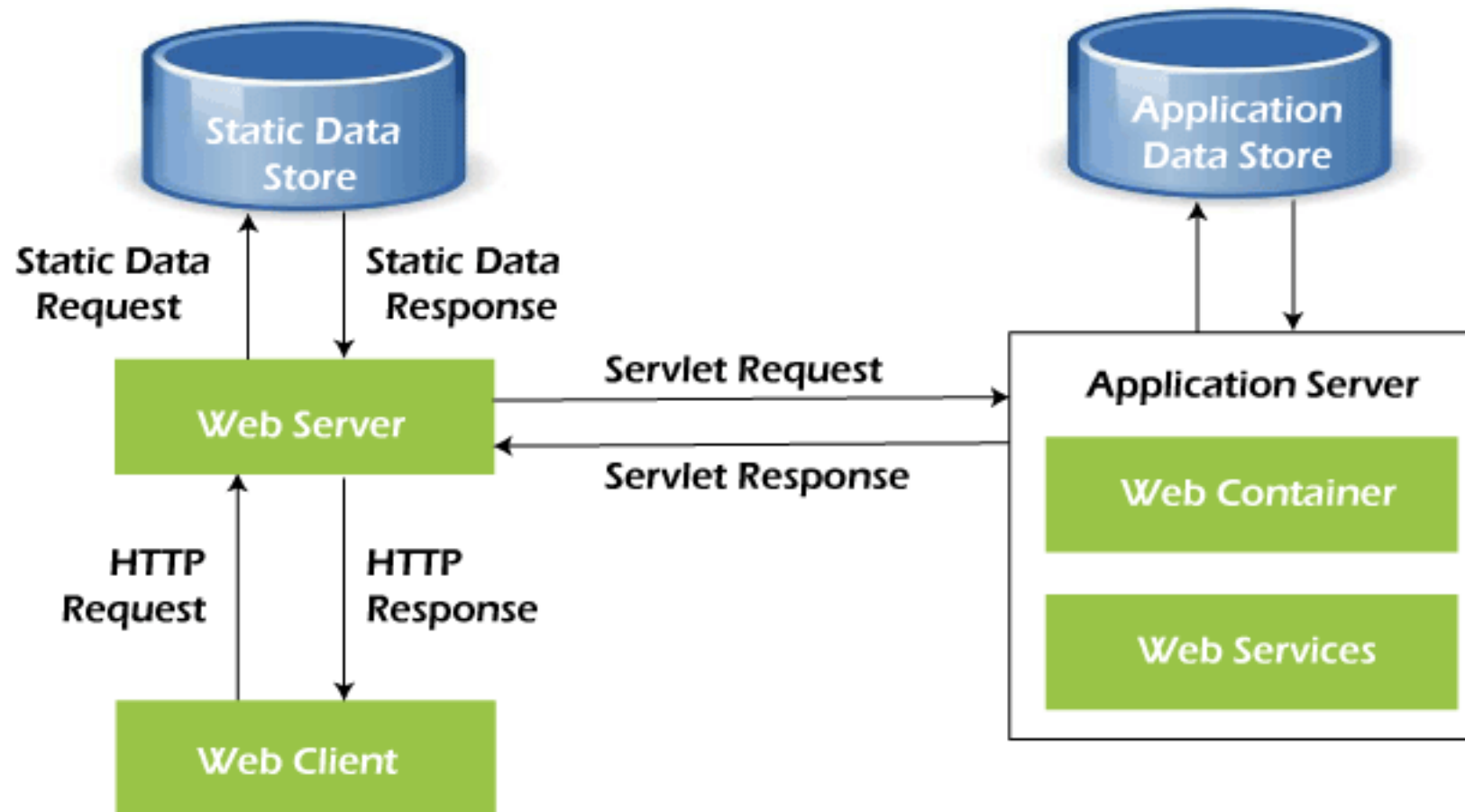


02

Servidores web y NGINX

Servidor web

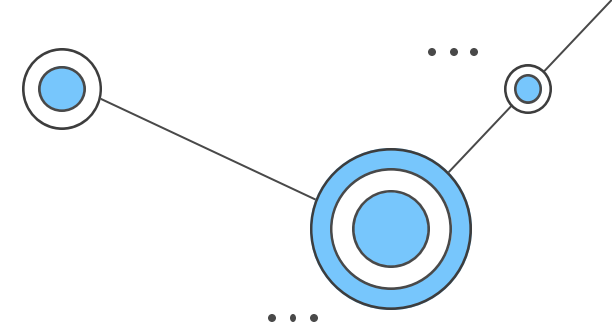
Working of web servers



Diversos servidores “autocontenidos”



Nginx



Rápido

El más rápido vs sus tres principales competidores

Loop de eventos

Facilidad para proveer miles de conexiones



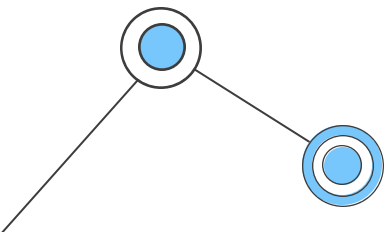
NGINX

Bien establecido

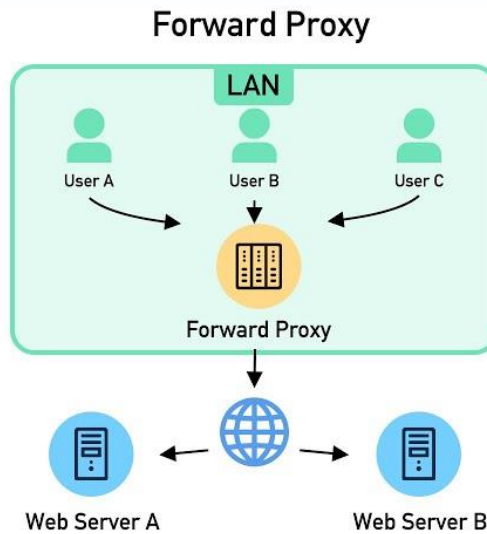
Años probado en producción

Flexible y extensible

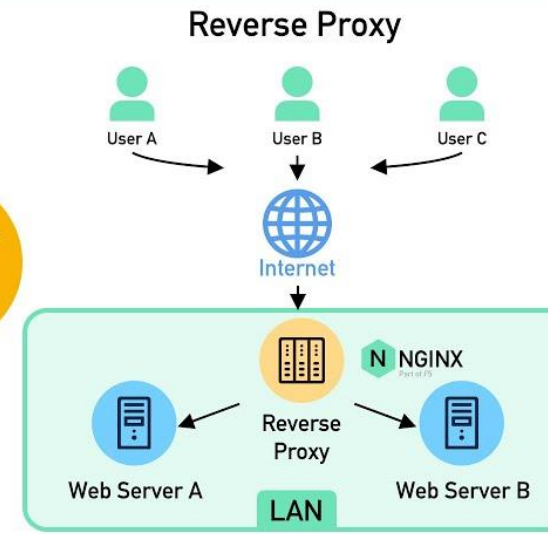
Diversas extensiones (lua) y capacidades



Forward Proxy vs. Reverse Proxy

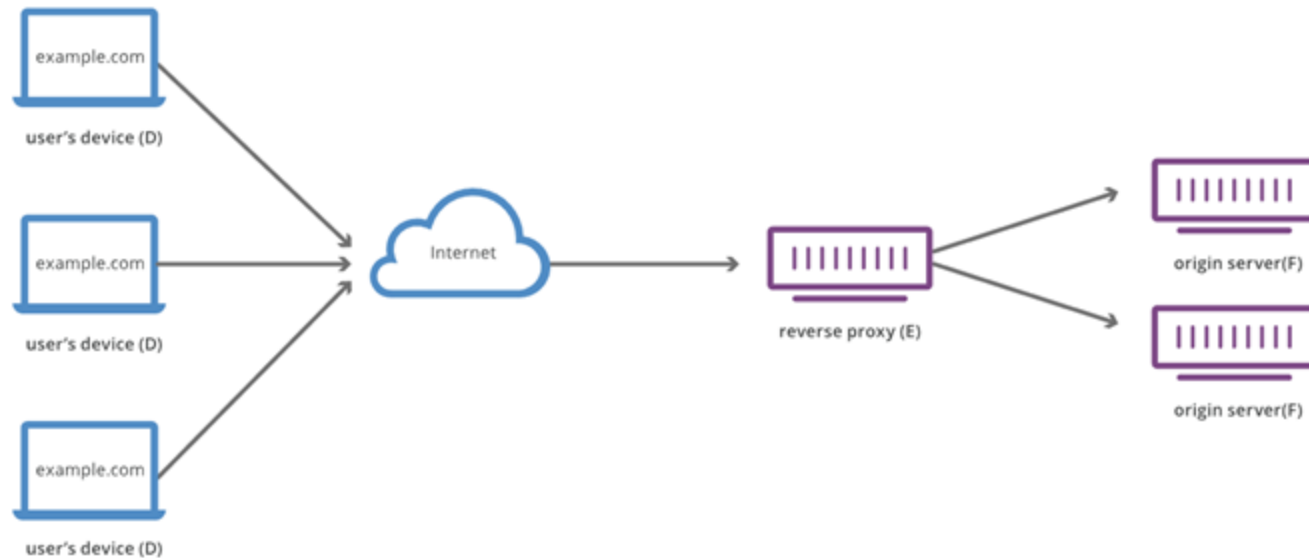


VS



Reverse proxy

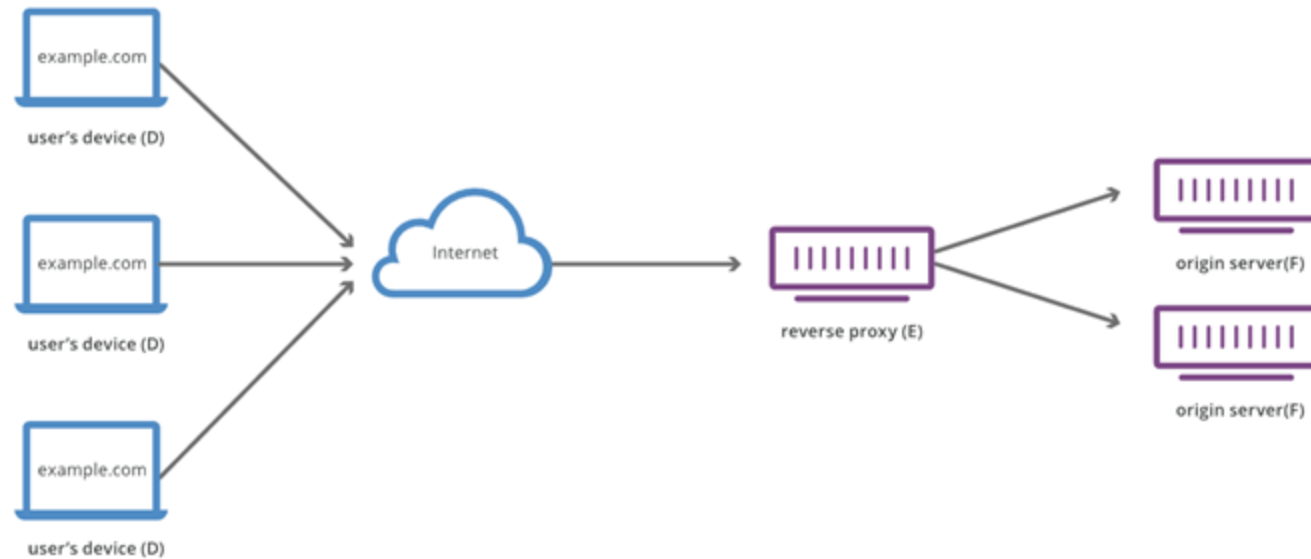
Reverse Proxy Flow



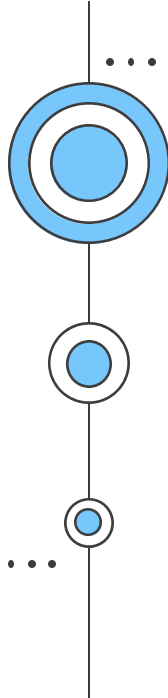
- Intermediario entre un usuario y un servidor web
- Permite implementar soluciones arquitectónicas (e.g. API gateway)
- Puede agregar balanceo de carga
- Protege ante ataques
- Cache
- Encriptación SSL

SSL y Load Balancer

Reverse Proxy Flow

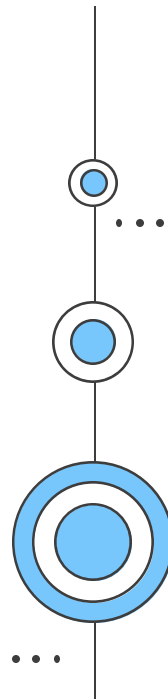


- Puede agregar balanceo de carga
- Encriptación SSL

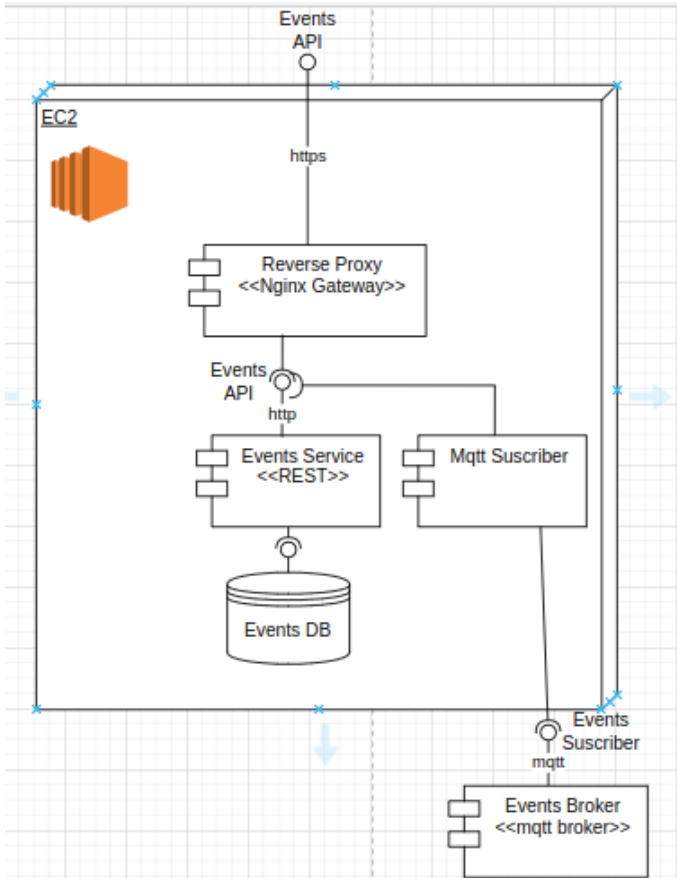


03

Deployment básico

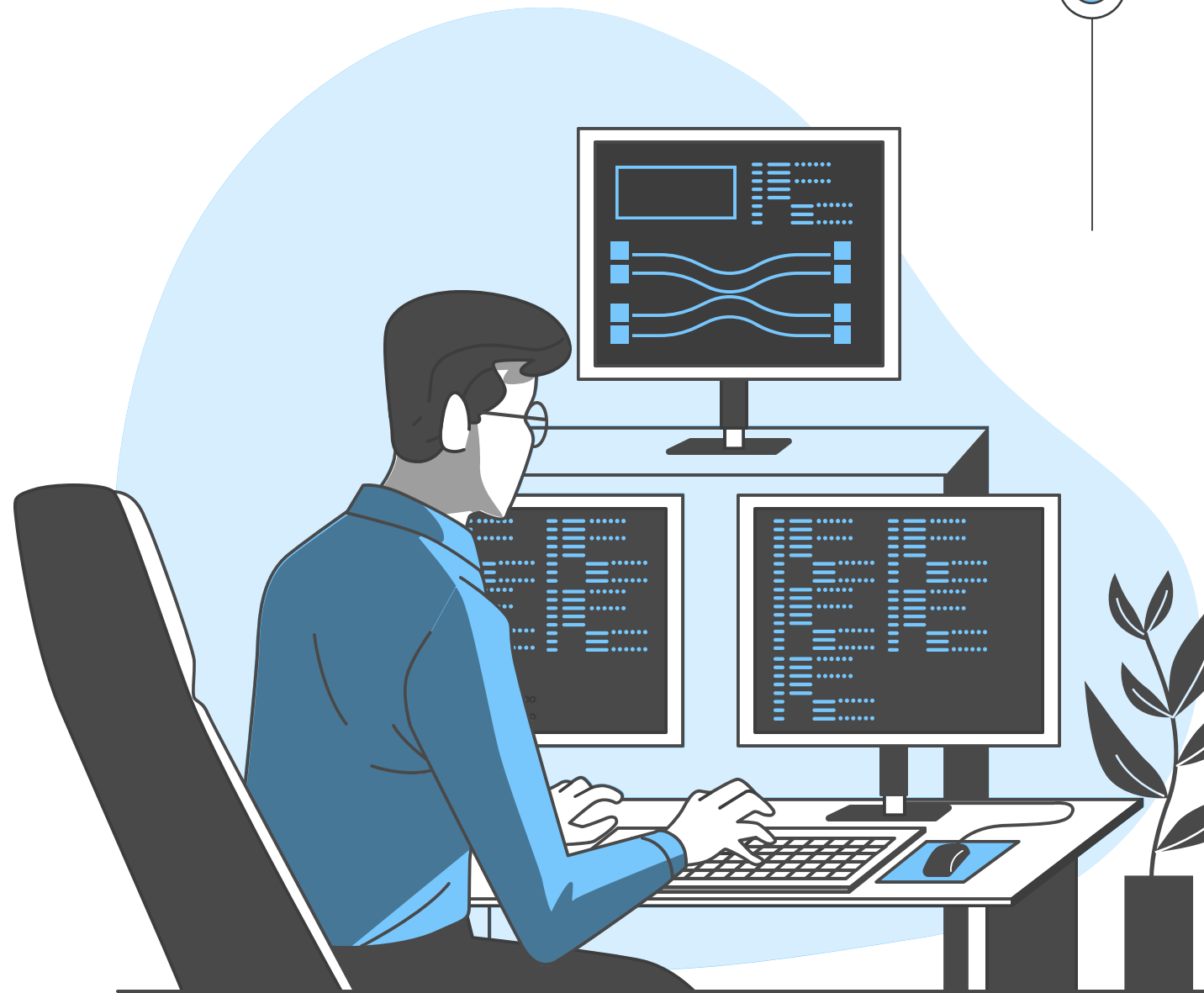


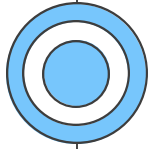
Pasos clave



- Tener un servidor levantado y una forma de mapear accesos al puerto 80 (API en puerto 80 directamente o usar Nginx)
- Conectarse al servidor
- Crear una clave ssh en el servidor y agregar esta clave a Github ([Referencia](#))
- Clonar el repositorio en el servidor
- Instalar dependencias necesarias (Docker)
- Configurar Nginx
- Correr la aplicación desde la carpeta del repositorio

Veámoslo en la práctica





Ayuda para instalar docker en EC2

1. Actualizar paquetes

```
sudo apt update
```

```
sudo apt upgrade -y
```

2. Instalar dependencias necesarias

```
sudo apt install -y ca-certificates curl gnupg lsb-release
```

3. Agregar la clave oficial de Docker

```
sudo mkdir -p /etc/apt/keyrings
```

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
```

```
sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
```

4. Agregar el repositorio oficial

```
echo \
```

```
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
```

```
https://download.docker.com/linux/ubuntu \
```

```
$(lsb_release -cs) stable" | \
```

```
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

5. Actualizar nuevamente

```
sudo apt update
```

6. Instalar Docker + Compose

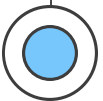
```
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

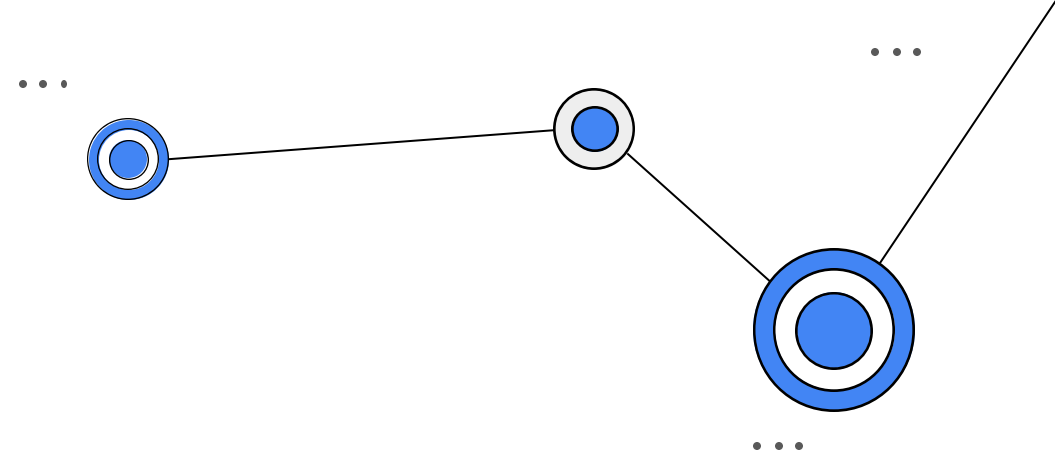
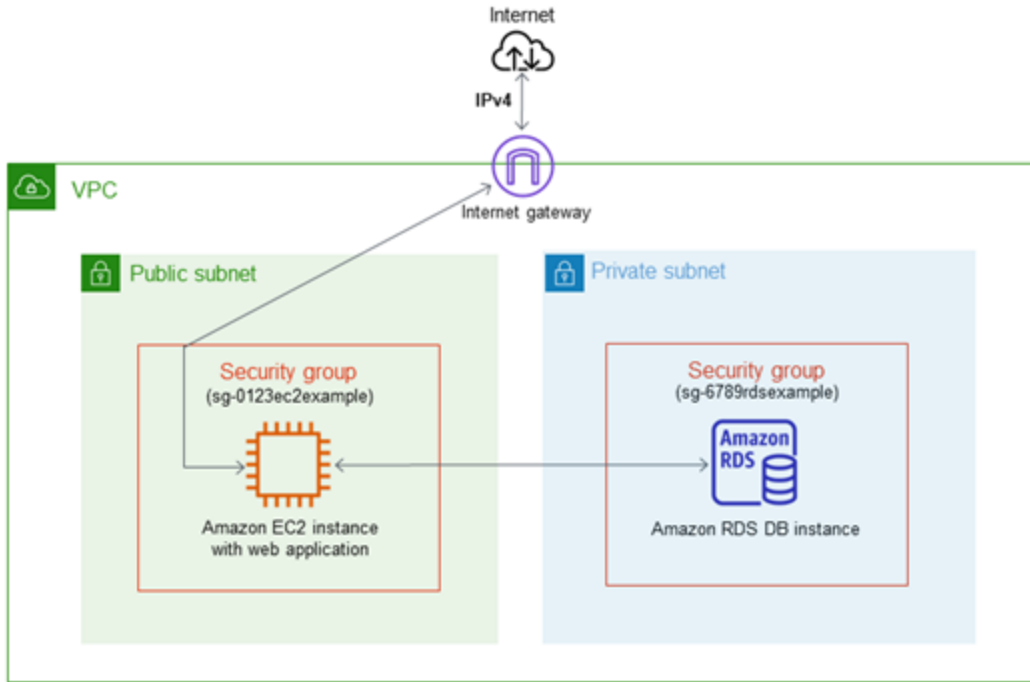
7. Verificar instalación

```
docker --version
```

```
docker compose version
```

```
sudo usermod -aG docker $USER
```





DNS, Nginx y Deployment

IIC2173 – 2025.2

DNS RECORDS CHEAT SHEET - CONSTELLIX

A (address)

1

A (address) - Most commonly used to map a fully qualified domain name (FQDN) to an IPv4 address and acts as a translator by converting domain names to IP addresses. ✓

AAAA (quad A)

2

AAAA (quad A) - Similar to A Records but maps to an IPv6 address (smart-phones prefer IPv6, if available). ✓

ANAME

3

ANAME - This record type allows you to point the root of your domain to a hostname or FQDN. ✓

CNAME

4

CNAME (Canonical Name) - An alias that points to another domain or subdomain, but never an IP address. Alias record mapping FQDN to FQDN, multiple hosts to a single location. This record is also good for when you want to change an IP address over time as it allows you to make changes without affecting user bookmarks, etc. ✓

SOA (start of authority)

5

SOA (Start of Authority) - Stores information about domains and is used to direct how a DNS zone propagates to secondary name servers. ✓

NS (name server)

6

NS (name server) - Specifies which name servers are authoritative for a domain or subdomains (these records should not be pointed to a CNAME). ✓

MX (mail exchange)

7

MX (Mail eXchange) - Uses mail servers to map where to deliver email for a domain (should point to a mail server name and not to an IP address). ✓

TXT (text)

8

TXT (text) - Allows administrators to add limited human and machine-readable notes and can be used for things such as email validation, site, and ownership verification, framework policies, etc., doesn't require specific formatting. ✓

SRV (service)

9

SRV (service) - Allows services such as instant messaging or VoIP to be directed to a separate host and port location. ✓

SPF (sender policy framework)

10

SPF (sender policy framework) - Helps prevent email spoofing and limits spammers. ✓

PTR (pointer)

11

PTR (pointer) - A reverse of A and AAAA records, which maps IP addresses to domain names. These records require domain authority and can't exist in the same zone as other DNS record types (put in reverse zones). ✓

QUICK TIP

12

Tip: Always check for typos and mistakes when entering your DNS record information, especially your IPs. The Zone Config File is a good place to check your work and spot any mistyped information. ✓

